

[illegible]

1. Cold-air snowball fight

WHAT YOU SEE

Children hurling snowballs in a frozen winter yard — the cold, dry air whipping past is the trigger: it dries and chills the airway lining and sets off reflex bronchoconstriction.

WHAT IT ENCODES

Cold, dry air is a classic asthma trigger: rapid heat and water loss from the airway mucosa during hyperventilation activates mast cells and sensory nerves, causing reflex bronchoconstriction. It frequently combines with exercise and viral season in winter to provoke attacks.

2. Exercise / running kids

WHAT YOU SEE

Kids sprinting and sledding hard in the snow — vigorous exertion in cold air is the visual for exercise-induced bronchoconstriction.

WHAT IT ENCODES

Exercise-induced bronchoconstriction (EIB) peaks 5–15 minutes after exertion (worse in cold/dry air) from airway dehydration and osmotic mast-cell activation, then resolves over 30–60 minutes. Prevention: a SABA 10–15 min before exercise, or — preferred long-term — regular ICS or ICS-formoterol, plus warm-up and an LTRA option.

BOARD TRAP

Treat EIB by optimizing CONTROLLER therapy (ICS/ICS-formoterol) rather than relying on daily/chronic SABA — frequent SABA need before exercise signals inadequately controlled underlying asthma. Isolated EIB in an athlete may need confirmation by exercise/eucapnic challenge.

3. Fireball / shortness of breath

WHAT YOU SEE

A flaming fireball/comet with 'shortness of breath' annotation.

WHAT IT ENCODES

Smoke (tobacco and wood/biomass), fumes, occupational irritants, and ambient air pollution (ozone, particulates) are environmental triggers that inflame airways and precipitate acute dyspnea and exacerbations. Pollution spikes correlate with ED visits for asthma.

4. Snowman with cannon (airway obstruction)

WHAT YOU SEE

A child aiming a toy cannon at the snowman — the cannon barrel is a narrowed, spasming airway, the visual for the acute airflow obstruction at the heart of an attack.

WHAT IT ENCODES

The central event of an exacerbation is acute, largely reversible airflow obstruction from bronchial smooth-muscle spasm, mucosal edema, and mucus plugging — producing wheeze, prolonged expiration, air-trapping/hyperinflation, and a falling PEF/FEV1.

5. Blue monster (hypoxia / cyanosis)

WHAT YOU SEE

A big BLUE monster looming in the foreground — the blue color literally IS cyanosis from severe hypoxemia during a bad exacerbation.

WHAT IT ENCODES

Severe exacerbation causes ventilation-perfusion mismatch and hypoxemia (cyanosis, falling SpO2). Early attacks show a respiratory ALKALOSIS from hyperventilation (low pCO2); as the patient tires, pCO2 rises toward and above normal — a normal/rising pCO2 is a red flag for impending respiratory failure.

BOARD TRAP

Cyanosis, silent chest, exhaustion, drowsiness, and a 'normalizing' or rising pCO2 are LATE, life-threatening signs — escalate to high-flow O2, continuous bronchodilators, IV steroids/magnesium, and ICU/intubation rather than reassuring on a 'normal' gas.

6. Spilled bricks/blocks (inflammation)

WHAT YOU SEE

Scattered red blocks/bricks dumped on the snow — the red rubble is airway inflammation (mucosal edema and mucus plugging) building up, the reason steroids are needed, not bronchodilators alone.

WHAT IT ENCODES

Beyond bronchospasm, exacerbations involve airway inflammation with mucosal edema and tenacious mucus plugging (sometimes Curschmann's spirals and Charcot-Leyden crystals on sputum), which is why systemic corticosteroids — not bronchodilators alone — are essential to resolve an attack.

7. Snowballs as allergens/triggers

WHAT YOU SEE

Piles of round snowballs being stockpiled as ammunition — each snowball is an inhaled aeroallergen (dust mite, pollen, dander) ready to pelt the airway and set off IgE-mediated attacks.

WHAT IT ENCODES

Aeroallergens — house dust mite, pollen, pet dander, cockroach, and mold — are common provoking triggers in atopic asthma, driving IgE-mediated mast-cell degranulation. Allergen avoidance and, in selected cases, immunotherapy or anti-IgE therapy reduce exacerbations.

8. Heart / red gift (cardiac mimic)

WHAT YOU SEE

A red heart-shaped object among the foreground figures — the heart flags the cardiac mimic: 'cardiac asthma' wheeze from left heart failure, not true asthma.

WHAT IT ENCODES

'Cardiac asthma' — wheezing from pulmonary congestion in left heart failure — can mimic an asthma exacerbation, especially in older patients. Clues favoring HF: orthopnea, paroxysmal nocturnal dyspnea, peripheral edema, raised JVP, S3, bibasilar crackles, and elevated BNP.

BOARD TRAP

New-onset wheeze in an ELDERLY patient with cardiac risk factors may be heart failure, not asthma — treating it with bronchodilators alone (and missing diuresis) is the wrong move.

9. Green sled / toy (viral trigger)

WHAT YOU SEE

A green sled/figure being pushed through the snow — the wintry sledding scene is the visual for viral URIs (rhinovirus, RSV, flu), the single most common exacerbation trigger and the cause of the winter peak.

WHAT IT ENCODES

Viral upper-respiratory infections (rhinovirus most commonly, also RSV, influenza) are the single most frequent trigger of asthma exacerbations across all ages, accounting for the majority of attacks. This drives the winter peak in asthma hospitalizations.

BOARD TRAP

Most exacerbations are VIRAL — antibiotics are not routinely indicated and should be reserved for clear evidence of bacterial infection (e.g., fever with purulent sputum and consolidation). Annual influenza vaccination is recommended.

10. Snowmen building (chronic remodeling)

WHAT YOU SEE

Two finished snowmen standing rigid in the yard — built up layer by layer and frozen stiff, they are the visual for chronic airway remodeling: a fixed, hardened structure that no longer melts away with bronchodilators.

WHAT IT ENCODES

Repeated or chronically uncontrolled inflammation leads to airway remodeling — subepithelial fibrosis, smooth-muscle hypertrophy, goblet-cell hyperplasia, and angiogenesis — producing a partially fixed obstructive component that does not fully reverse with bronchodilators.

11. Warm tent / shelter (treatment refuge)

WHAT YOU SEE

A warm covered porch/tent with seated figures on the right — the refuge from the cold is the visual for acute exacerbation treatment: oxygen, bronchodilators, and early systemic steroids.

WHAT IT ENCODES

Acute exacerbation management: controlled O₂ to target SpO₂ 93–95% (88–92% if at COPD-overlap risk), repeated/continuous inhaled SABA (± ipratropium for severe), and early SYSTEMIC corticosteroids (oral prednisolone ~40–50 mg ×5–7 days, equal to IV). Add IV magnesium sulfate for severe/refractory attacks; consider escalation to NIV/intubation if failing.

12. Person resting indoors (recovery)

WHAT YOU SEE

A figure resting comfortably inside the warm shelter — recovery is the visual cue to step UP controller therapy and review technique, not just relieve symptoms.

WHAT IT ENCODES

Recovery is not just symptom relief: every exacerbation mandates a review of inhaler technique, adherence, triggers, and a written action plan, and a STEP-UP of controller therapy (e.g., start or increase ICS / move to ICS-formoterol MART) to reduce future risk. Arrange follow-up within 2–7 days.

BOARD TRAP

Don't treat the acute attack and discharge on SABA alone — failing to start/intensify ICS controller therapy after an exacerbation is a key board error and a driver of recurrent attacks and asthma death.